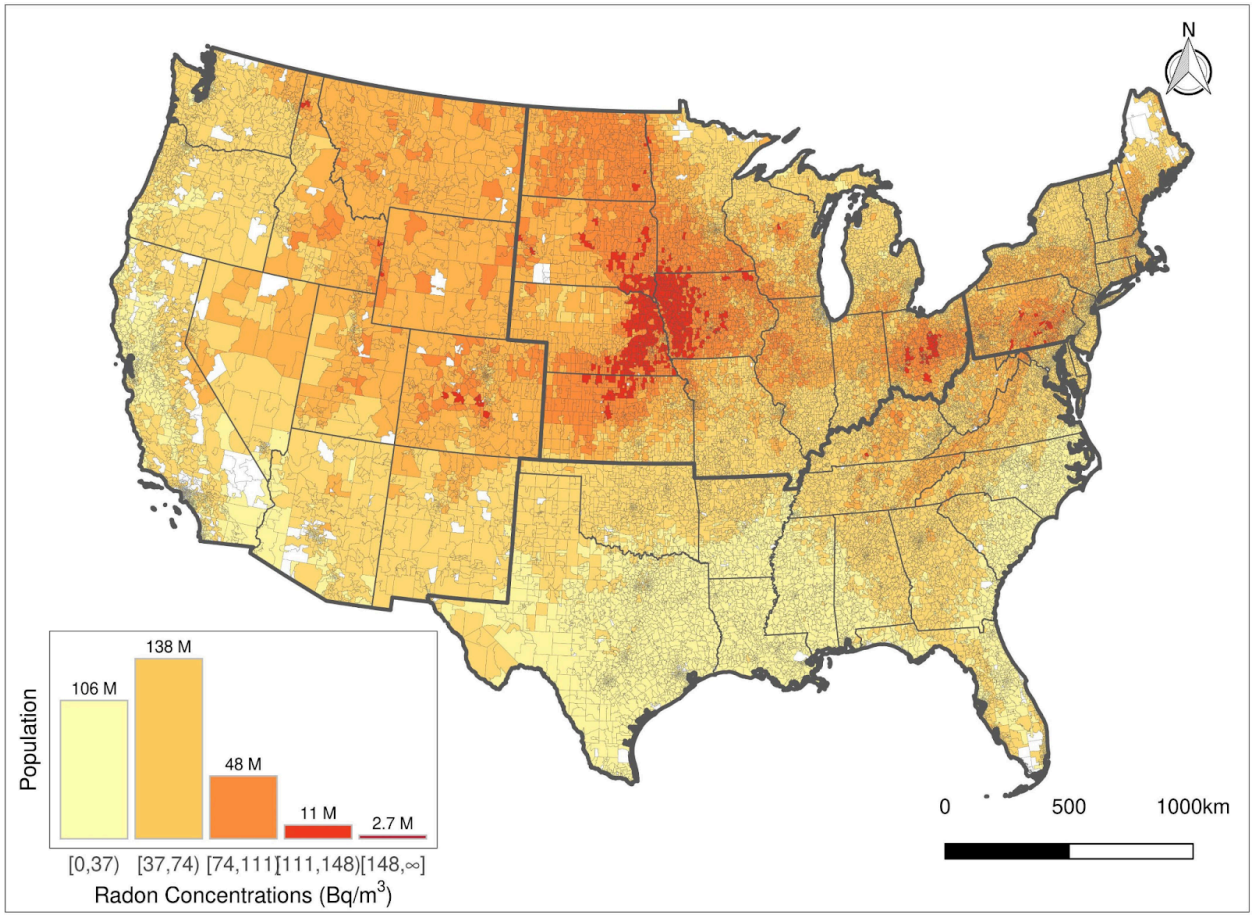
Group 10

Radon Mitigation Project Report Pt 1



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I Project Description

1 Project Overview

Radon Mitigation is a decentralized data collection and distribution solution to reduce the impact of radon on home residents.

2 The Purpose of the Project

With Radon Mitigation, People will be able to see local radon levels, knowing the radon levels within the property area, around construction sites. Users will be able to get Lifetime Radiation exposure estimates and risks levels at easy access. They will gain better awareness of the amount of radon they are currently exposed to on their property or within their area they are in.

2a The User Business or Background of the Project Effort

In the Real Estate Business, The client typically is most likely to do radon testing when they are buying or selling a home. Currently when they do this testing they use radon test kits which can detect high levels of radon.

2b Goals of the Project

This project will help users have easy access to data on radon levels around their property/ current area and get a better understanding on the current exposure they have to radon along with their current risk level.

2c Measurement

We will know when the product overall has improved for our clients when data shows that people with the app have more awareness of the radon levels on their property and possibly have decreased exposure to radon. We can measure this simply by using our data to see how many people are using our product. The more people using our product means thats more people who now have awareness on what radon is and the current exposure they have or could potentially have to it.

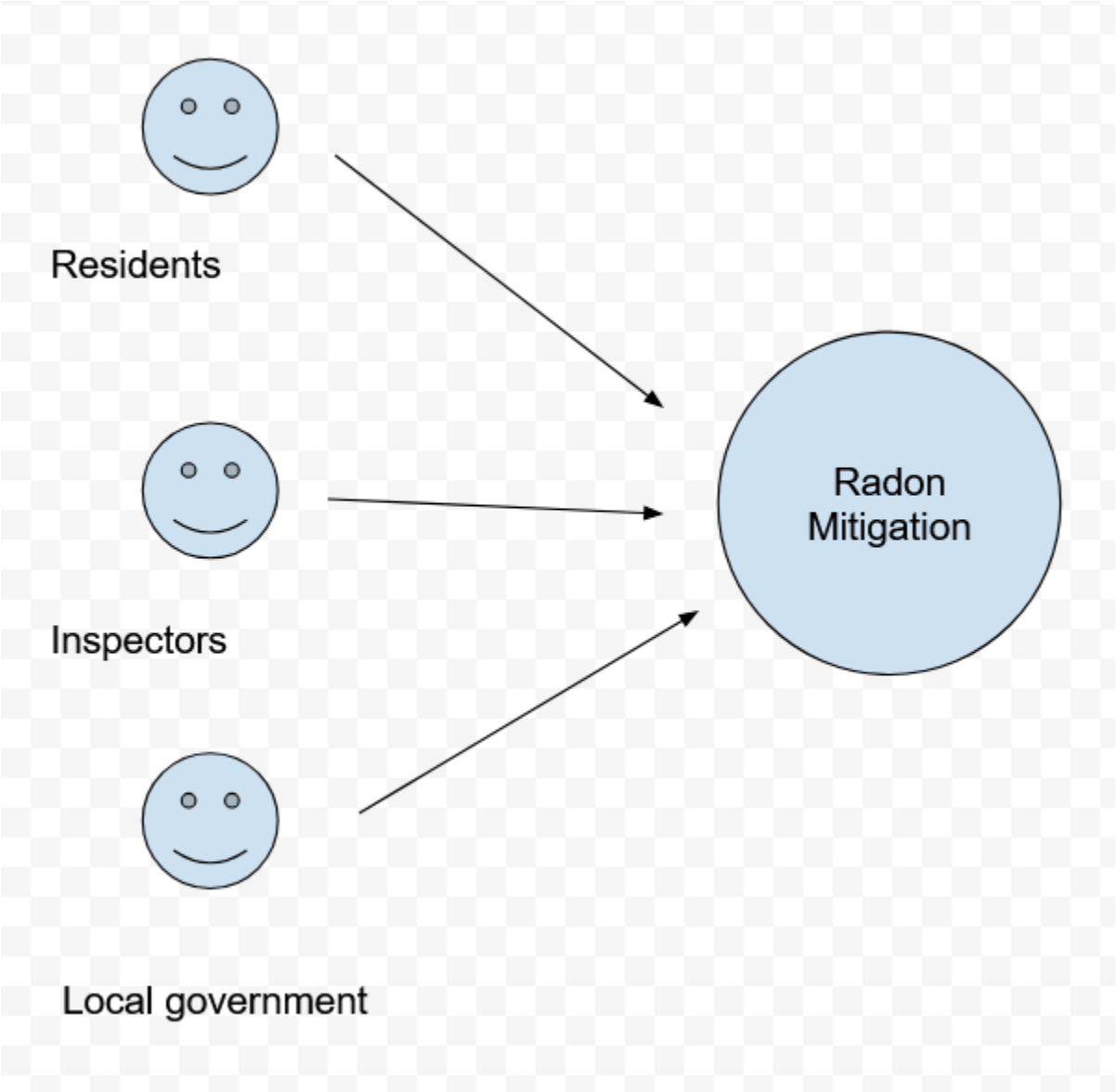
3 The Scope of the Work

Our product will provide local radon levels information given by authorized users. Along with allowing users to store new radon information and display all this information and exposure estimates onto the platform

3a The Current Situation 3b The Context of the Work

Currently, homeowners are required to hire licensed professors to do radon testing. That info is reported most to the state government which keeps that data to be used for state/ country level mapping of radon levels.

3b The Context of the Work



Everyone can read and verified users can write

3c Work Partitioning

Event Name	Input	Output	Summary
User Creates an account	Username, Email, and Password	Username, Email, and Password	Username, Email, and Password

User logs into account	Username/Email and password	Successfully logged in	System checks if username and password exist in system and logins user in if it does
User searches map for local area radon	User types in address in map search bar	Area of the address with radon levels and estimates shown	Map will pan over to the address the user typed in and shows the area radon levels by numbers and colors to represent how high the level is and it displays estimates
User inputs new radon info	Address, radon levels, calculator, valid photo for accuracy	Successfully submitted radon info. Data will be reviewed for validation	The system will store the users input info and will validate that the user is verified and have some one check to make sure all info is valid before officially posting it on website

3d Competing Products

https://iemaohs.illinois.gov/nrs/radon/radonillinois.html

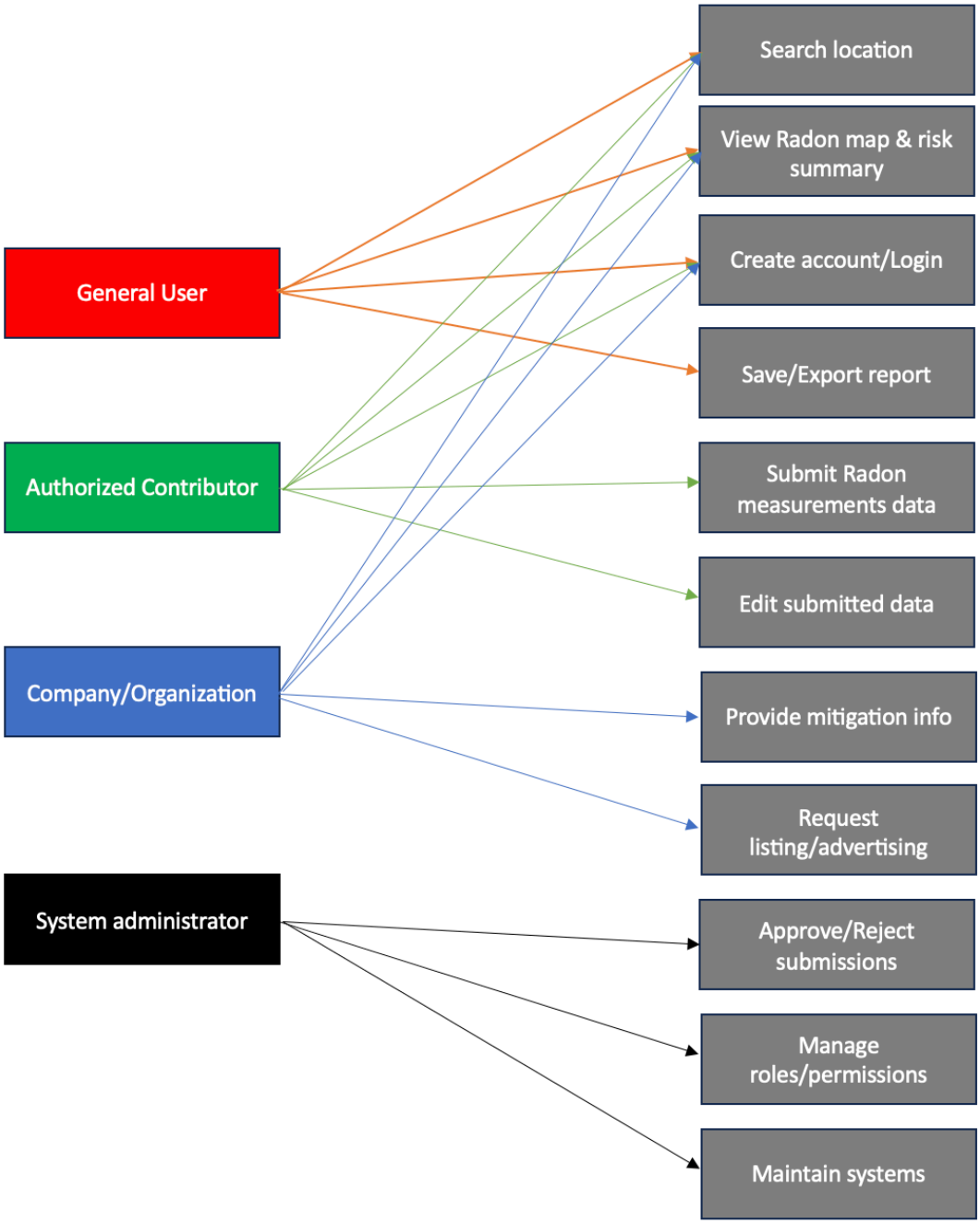
This website gives radon levels of city neighborhoods/ zip codes. However most of the data is dated between 2003 - 2019 and the data includes *"only the highest measurement for each address that was submitted*". Our product would contain all radon levels to be found in a visual map along with having more recent up-to-date information unlike the current Illinois website only having data up to 2019.

4 The Scope of the Product

The platform is designed to provide location based Radon exposure information through a centralized platform that aggregates environmental and user submitted data. This platform allows users to view Radon levels by geographic area, contribute to measurement data as

authorized and access insights related to potential Radon exposure risks. The platform itself doesn't perform physical Radon measurements, but instead manages data collection, visualization and analysis. External actors including general users, authorized contributors and organizations utilize the platform through role based access, while the system handles data processing, mapping and reporting data.

4a Scenario Diagram(s)



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4b Product Scenario List

Scenerio	External Actors	Description
View Radon Levels by Location	General User	User searches an address/area to view Radon exposure levels
Submit Radon Measurement Data	Authorized Contributor	Contributor uploads radon test results to the platform
Company Information Display	Company/Organization	Organization inputs mitigation system information
Account Creation & Access	General User	Users create accounts to access reports or contribute data depending on permissions.
System Maintenance & Updates	System Administrator	Administrator manages database, security & updates

4c Individual Product Scenarios

1. Viewing Radon Levels by Location

User accesses the platform through a web browser and enters a home address or geographic location into the search bar. The system finds and retrieves available Radon data from the database and displays the information on an interactive map along with approximate estimated exposure levels. The user reviews the results to understand potential risks of Radon in that area.

2. Submitting Radon Measurement Data

Authorized contributor logs into the platform using verified credentials. The contributor selects a property location and uploads radon measurement results obtained during testing. The system validates the input and stores the data in the central database, updating the visualization for future users.

3. Company Information Display

Company representatives log into the platform and submit information about their mitigation services or system specifications. Once approved, the company input information becomes visible to users who are searching for Radon related queries in relevant areas.

4. Account Creation & Access

Users create accounts by providing basic registration information. Post registration, the user can access additional features such as saved data reports, notifications or contribution options depending on permissions.

5. System Maintenance & Updates

System administrator accesses internal backend tools to manage server performance, update databases, monitor security pipelines and update software to ensure the platform operates correctly.

5 Stakeholders

The stakeholders in this application will be home owners, realtors, contractors, and Radon Mitigation companies.

5a The Client

We do not have a particular client. Since we are developing a platform, initially we will be the ones to decide how we want to do things. From internal algorithms to public facing UI, we have full control over what this application does. Once deployed however, we will rely on feedback and reviews from our most loyal users to help develop a better app.

5b The Customer

We do not have a customer. We will at some point display ads to generate an 'enough' stream of revenue for upkeeping and paying salaries. We do not intend to steal data, and so this application will be limited by its algorithmic potential.

5c Hands-On Users of the Product

Anyone can be a user of this product. Anyone that wants to view/add to the Radon database can use this product. One may create an account to retrieve radon reports 1 time, or come again and again to add/modify to the database.

5d Maintenance Users and Service Technicians

This does not apply to our project. Because this is a website also designed for mobile devices, users simply enter the url to get to it. We (our team) will be the ones responsible for

creating and upkeep the server where the website is hosted. We will do server management, database optimization, and security patches over the life of the product and our company.

5e Other Stakeholders

Sometimes the government likes to 'adopt' a certain project and give some financial aid to help a cause. There could be a chance this app gets some recognition from the gov.

Another example would be if a particular radon company doesn't like what we are doing. Maybe they oversell and underdeliver, or lie about how much radon is actually showing up on a property, in that case, they would hate to see this product.

5f User Participation

Users may report bugs, other users, or provide feedback. We will have a clear and easy way to leave feedback once the initial version of the product is ready.

5g Priorities Assigned to Users

Normal User - they simply can read the values given an address or street

Contractors - they can read as well as update the radon levels from a certain address. To get this level of access, they must sign contracts, and get certified.

Companies - they can read as well as input their products and specifications. They can request advertisements of their products by paying money. They must go through an approval process.

6 Mandated Constraints

6a Solution Constraints

The system must functionally work as a platform that allows users to view Radon levels by geographical locations whilst also allowing authorized contributors to submit measurement data regarding Radon levels. The system will present Radon levels using standard measurement units(picocuries/liter), consistent with public health guidelines. Since this platform relies on aggregated environmental and user submitted data, mechanisms must be implemented to ensure data validation, reliability and controlled modification. The application must also remain accessible across consumer devices without the requirement of any specialized hardware.

6b Implementation Environment of the Current System

The platform shall operate as a web based platform that can be accessed through modern desktop/mobile browsers across multiple operating systems such as Windows, macOS, Android and iOS, without requiring additional software installation. The backend will be hosted on cloud servers for data storage, geospatial processing & analytics and security with internet connectivity required for accessing maps and submitting/retrieving Radon data.

The system will additionally integrate with external sources such as publicly available geological datasets, environmental records and mapping APIs to visualize Radon levels by location, with potential future integrations to improve accuracy and coverage.

6c Off-the-Shelf Software

Commercial off-the-shelf components may be used for cloud hosting, database management, mapping services and application development frameworks. These tools will provide foundational functionalities such as user authentication, geographic visualization and secure data storage without having to custom develop these systems from scratch.

6d Anticipated Workplace Environment

Users can access the system in a variety of different environments including homes, offices, outdoor locations, etc wherever the user has access to active internet connectivity. Some users may interact with the platform while performing construction activities and inspections to input Radon data into the platform. Thus the interface must be responsive and optimized for both mobile and desktop usage.

6e Schedule Constraints

Project timelines will primarily be determined by phase wise internal development milestones. Initial prototypes shall be completed early to demonstrate feasibility, followed by continuous iterative development cycles, which shall also be integrated with cyclical testing phases. Integration of external datasets or new features may introduce dependencies that could also influence the schedule.

6f Budget Constraints

The project will initially operate with limited financial resources with only usage of efficient development tools and infrastructure. Costs include cloud hosting services, data storage, mapping APIs and maintenance resources. Budgets may inflate in order to account for any further functionality & technology changes during further development stages.

7 Naming Conventions and Definitions

7a Definitions of Key Terms

Radon: a colorless, odorless, and tasteless radioactive gas produced by the natural decay of uranium and radium in soil, rock, and groundwater

Mitigation: the action of reducing the severity, seriousness, or painfulness of something.

Picocuries: a minute unit of radioactivity, representing one trillionth of a Curie. It measures the rate of radioactive decay, with 1 pCi equaling approximately 2.22 disintegrations per minute.

API: a set of rules and protocols that allows different software applications to communicate, exchange data, and share functionality

7b UML and Other Notation Used in This Document

This document includes UML diagrams that follows the Version 2.0 OMG UML standard, as outlined by Fowler in UML Distilled, Third Edition, Boston: Pearson Education, 2004.

7c Data Dictionary for Any Included Models

New Radon data: Address, city, zipcode, Radon levels in units, date data was recorded, Picture for valid proof of radon data

Account Details: username, email address, password, and profile picture (optional).

8 Relevant Facts and Assumptions

8a Facts

- The US EPA estimates the average indoor radon concentration is 1.3 picocuries per liter (pCi/L).
- It's estimated approximately 1 in 15 homes in the US has high radon
- Radon comes from the soil, water, and water supply. Some building materials also release radon, such as concrete, granite counters, and wall boards
- Given that Radon is often reduced easily and inexpensively once the threat is known by currently using test kits, the same effect should occur by similar finding out the exact same information using our product

8b Assumptions

- All users will be using computers that can open websites.
- User will need internet connect so that the product can connect to database
- Its assumed that users at least have a very basic knowledge on what radon is